

§ 1.10.

P1. $y' = 2t(y+1), y(0) = 0$

$$f(t, y) = 2t(y+1)$$

$$\Rightarrow y(t) = y_0 + \int_{t_0}^t f(s, y(s)) ds$$

$$= \int_0^t 2s(y(s)+1) ds$$

$$y_1(t) = \int_0^t 2s(y_0+1) ds = \int_0^t 2s ds = t^2$$

$$y_2(t) = \int_0^t 2s(y_1(s)+1) ds = \int_0^t 2s(s^2+1) ds = t^2 + \frac{t^4}{2!}$$

$$y_3(t) = \int_0^t 2s(y_2(s)+1) ds = \int_0^t 2s(1+s^2+\frac{s^4}{2}) ds = t^2 + \frac{t^4}{2!} + \frac{t^6}{3!}$$

...

$$y_n(t) = \int_0^t 2s(y_{n-1}(s)+1) ds = \int_0^t 2s \cdot (1+s^2+\frac{s^4}{2!}+\dots+\frac{s^{2(n-1)}}{(n-1)!}) ds = t^2 + \frac{t^4}{2!} + \frac{t^6}{3!} + \dots + \frac{t^{2n}}{n!}$$

clearly,

$$\lim_{n \rightarrow \infty} y_n(t) = \lim_{n \rightarrow \infty} (t^2 + \frac{t^4}{2!} + \frac{t^6}{3!} + \dots + \frac{t^{2n}}{n!}) = \lim_{n \rightarrow \infty} (t^2 + \frac{(t^2)^2}{2!} + \frac{(t^2)^3}{3!} + \dots + \frac{(t^2)^n}{n!}) = e^{t^2} - 1$$

P2. $y' = t^2 + y^2, y(0) = 1$

$$f(t, y) = t^2 + y^2$$

$$\Rightarrow y(t) = y_0 + \int_{t_0}^t f(s, y(s)) ds$$

$$= 1 + \int_0^t (s^2 + y(s)^2) ds$$

$$y_1(t) = 1 + \int_0^t (s^2 + y_0^2) ds = 1 + \int_0^t (1 + s^2) ds = 1 + t + \frac{t^3}{3}$$

$$y_2(t) = 1 + \int_0^t (s^2 + y_1(s)^2) ds = 1 + \int_0^t (s^2 + (1+s+\frac{s^3}{3})^2) ds = 1 + \int_0^t (1 + 2s + 2s^2 + \frac{2}{3}s^3 + \frac{2}{3}s^4 + \frac{s^6}{9}) ds = 1 + t + \frac{t^2}{2} + \frac{2}{3}t^3 + \frac{1}{6}t^4 + \frac{2}{15}t^5 + \frac{1}{63}t^7$$

P4. Solution:

Here $t_0 = 0, y_0 = 1, a = \frac{1}{2}$. Let R be the rectangle $0 \leq t \leq \frac{1}{2}, |y| \leq b$.

$$M = \max_{(t,y) \in R} |f(t, y)| = \max_{\substack{0 \leq t \leq \frac{1}{2} \\ |y| \leq b}} |y^2 + 6t^2|$$

$$= b^2 + 1$$

$$\alpha = \min(a, \frac{b}{M}) = \min(\frac{1}{2}, \frac{b}{b^2+1}) = \min(\frac{1}{2}, \frac{1}{b+\frac{1}{b}})$$

Let us choose $b = 1$ (i.e., $b = \frac{1}{b}$), then we get

$$\alpha = \min(\frac{1}{2}, \frac{1}{2}) = \frac{1}{2}$$

Hence, $y(t)$ exists on $0 \leq t \leq \frac{1}{2}$.

P5. Solution: Here $t_0 = 0, y_0 = 0, a = (\frac{1}{2})^{\frac{2}{3}}$

Let R be the rectangle $0 \leq t \leq (\frac{1}{2})^{\frac{2}{3}}, |y| \leq b$

$$M = \max_{(t,y) \in R} |f(t, y)| = \max_{\substack{0 \leq t \leq (\frac{1}{2})^{\frac{2}{3}} \\ |y| \leq b}} |t + y^2|$$

$$= (\frac{1}{2})^{\frac{2}{3}} + b^2$$

$$\alpha = \min(a, \frac{b}{M}) = \min((\frac{1}{2})^{\frac{2}{3}}, \frac{b}{(\frac{1}{2})^{\frac{2}{3}} + b^2}) = \min((\frac{1}{2})^{\frac{2}{3}}, \frac{1}{(\frac{1}{2})^{\frac{2}{3}} + b})$$

Let us choose $b = (\frac{1}{2})^{\frac{1}{3}}$ (i.e., $\frac{1}{b} = b$), we get

$$\alpha = \min((\frac{1}{2})^{\frac{2}{3}}, (\frac{1}{2})^{\frac{2}{3}}) = (\frac{1}{2})^{\frac{2}{3}}$$

Hence, $y(t)$ exists on $0 \leq t \leq (\frac{1}{2})^{\frac{2}{3}}$

P7. Solution: Here $t_0 = 0, y_0 = 0, a = \frac{1}{2}$

Let R be the rectangle $0 \leq t \leq \frac{1}{2}, |y| \leq b$.

$$M = \max_{(t,y) \in R} |f(t, y)| = \max_{\substack{0 \leq t \leq \frac{1}{2} \\ |y| \leq b}} |e^{-t^2} + y^2|$$

$$= 1 + b^2$$

$$\alpha = \min(a, \frac{b}{M}) = \min(\frac{1}{2}, \frac{b}{1+b^2}) = \min(\frac{1}{2}, \frac{1}{b+\frac{1}{b}})$$

Let us choose $b=1$, then we get

$$\alpha = \min\left(\frac{1}{2}, \frac{1}{2}\right) = \frac{1}{2}$$

Hence, $y(t)$ exists on $0 \leq t \leq \frac{1}{2}$.

P8. Solution: Here $t_0=1$, $y_0=0$, $a=\frac{\sqrt{e}}{2}$

Let R be the rectangle $1 \leq t \leq 1+\frac{\sqrt{e}}{2}$, $|y| \leq b$.

$$M = \max_{(t,y) \in R} |f(t,y)| = \max_{\substack{1 \leq t \leq 1+\frac{\sqrt{e}}{2} \\ |y| \leq b}} |e^{-t} + y^2|$$

$$= \frac{1}{e} + b^2$$

$$\alpha = \min\left(a, \frac{b}{M}\right) = \min\left(\frac{\sqrt{e}}{2}, \frac{b}{\frac{1}{e} + b^2}\right)$$

$$= \min\left(\frac{\sqrt{e}}{2}, \frac{1}{\frac{1}{eb} + b}\right)$$

Let us choose $b=\frac{1}{\sqrt{e}}$, then we get

$$\alpha = \min\left(\frac{\sqrt{e}}{2}, \frac{\sqrt{e}}{2}\right) = \frac{\sqrt{e}}{2}$$

Hence, $y(t)$ exists on $1 \leq t \leq 1+\frac{\sqrt{e}}{2}$

P9. Solution: Here $t_0=0$, $y_0=1$, $a=\frac{\sqrt{2}}{1+(1+\sqrt{2})^2}$

Let R be the rectangle $0 \leq t \leq \frac{\sqrt{2}}{1+(1+\sqrt{2})^2}$, $|y-1| \leq b$.

$$M = \max_{(t,y) \in R} |f(t,y)| = \max_{\substack{0 \leq t \leq \frac{\sqrt{2}}{1+(1+\sqrt{2})^2} \\ |y-1| \leq b}} |e^{-t} + y^2|$$

$$= 1 + (b+1)^2$$

$$\alpha = \min\left(a, \frac{b}{M}\right) = \min\left(\frac{\sqrt{2}}{1+(1+\sqrt{2})^2}, \frac{b}{1+(b+1)^2}\right)$$

$$= \min\left(\frac{\sqrt{2}}{1+(1+\sqrt{2})^2}, \frac{1}{\frac{2}{b} + b + 2}\right)$$

Let us choose $b=\sqrt{2}$ (i.e., $\frac{2}{b}=b$), we get

$$\alpha = \min\left(\frac{\sqrt{2}}{1+(1+\sqrt{2})^2}, \frac{\sqrt{2}}{1+(1+\sqrt{2})^2}\right) = \frac{\sqrt{2}}{1+(1+\sqrt{2})^2}$$

Hence, $y(t)$ exists on $0 \leq t \leq \frac{\sqrt{2}}{1+(1+\sqrt{2})^2}$

P14. Solution: Here $t_0=0$, $y_0=0$. Let R be the rectangle $0 \leq t \leq a$, $|y| \leq b$.

$$M = \max_{(t,y) \in R} |f(t,y)| = \max_{\substack{0 \leq t \leq a \\ |y| \leq b}} e^{-t} + \ln(1+y^2)$$

$$= 1 + \ln(1+b^2)$$

$$\alpha = \min\left(a, \frac{b}{M}\right) = \min\left(a, \frac{b}{1+\ln(1+b^2)}\right)$$

Notice that $\lim_{b \rightarrow \infty} \frac{b}{1+\ln(1+b^2)} = \infty$, so that we can make the quantity α as large as desired by choosing a and b sufficiently large. Therefore, $y(t)$ exists for $t \geq 0$.

$$P16. (a) M = \max_{(t,y) \in R} |f(t,y)| = \max_{\substack{0 \leq t \leq a \\ |y| \leq b}} |t^2 + y^2|$$

$$= a^2 + b^2$$

$$\alpha = \min\left(a, \frac{b}{M}\right) = \min\left(a, \frac{b}{a^2 + b^2}\right)$$

Therefore, $y(t)$ exists on $0 \leq t \leq \min\left(a, \frac{b}{a^2 + b^2}\right)$

(b). Let a be fixed.

$$\frac{b}{a^2 + b^2} = \frac{1}{\frac{a^2}{b} + b} \leq \frac{1}{2\sqrt{\frac{a^2}{b} \cdot b}} = \frac{1}{2\sqrt{a^2}} = \frac{1}{2a}.$$

where the equality is reached when $a=b$.

$$(c) \alpha = \min\left(a, \frac{1}{2a}\right)$$

$$= \begin{cases} a & \text{when } a \leq \frac{1}{2a} \\ \frac{1}{2a} & \text{when } a \geq \frac{1}{2a} \end{cases}$$

$$= \begin{cases} a & \text{when } a \leq \frac{1}{\sqrt{2}} \\ \frac{1}{2a} & \text{when } a \geq \frac{1}{\sqrt{2}} \end{cases}$$

$\Rightarrow \alpha \leq \frac{1}{\sqrt{2}}$ where the equality is reached when $a = \frac{1}{\sqrt{2}}$.

(d). It is natural to conclude that $y(t)$ exists on $0 \leq t \leq \frac{1}{\sqrt{2}}$ by choosing $a = \frac{1}{\sqrt{2}}$, $b = \frac{1}{\sqrt{2}}$ for the rectangle R .